R V R & J C COLLEGE OF ENGINEERING *(Autonomous)*,

Chowdavaram, Gunutr-19

B.Tech., Computer Science & Engineering (Data Science)

(w.e.f. the academic year 2021-2022)

Syllabus (R20) - Semester V (Third Year)

CD314 CDEL01	Data Engineering (Professional Elective- I)	L	T	P	C
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Course Objectives:

1. To introduce the basics of data warehousing and data mining.
2. To impart knowledge on data mining techniques.
3. To introduce mining on complex data objects.

Course Outcomes:

On completion of the course, the students will be able to:

1. Explain the concepts of data warehousing and data mining.
2. Extract association rules from transactional databases.
3. Extract association rules from transactional databases.
4. Build a classifier for a given data set.
5. Apply various clustering and outlier detection techniques for a given data set.
6. Describe the concepts of mining on complex data objects.

Course Content:

UNIT – I **[CO1, CO2]** **(13 Periods)**

Data Warehousing and Online Analytical Processing: Data Warehouse: Basic Concepts-Data Warehouse Modeling: Data Cube and OLAP-Data Warehouse Design and Usage- Data Warehouse Implementation.

Getting to know Your Data: Data Objects and Attribute Types- Basic Statistical Descriptions of Data- Measuring Data Similarity and Dissimilarity.

Data Preprocessing: An overview of Data Preprocessing- Data cleaning- Data Integration-Data Reduction- Data Transformation and Data Discretization.

UNIT – II **[CO1, CO3]** **(12 Periods)**

Introduction - Data Mining: Why Data Mining- What is Data Mining? -What Kinds of Data can be mined? - What Kinds of Patterns can be mined? - Which Technologies are used? - Major Issues in Data Mining.

Mining Frequent Patterns, Associations, and Correlations: Basic Concepts- Frequent Item set Mining Methods: Apriori Algorithm, Generating Association Rules, Improving the efficiency of Apriori, FP Growth Approach for Mining Frequent Item Sets, Mining Frequent Item Sets using Vertical Data Format Method.

UNIT – III **[CO4]** **(13 Periods)**

Classification: Basic Concepts- Decision tree induction- Bayes Classification Methods- Rule Based Classification- Model Evaluation and Selection- Techniques to Improve Classification Accuracy.

Advanced Methods in Classification: Bayesian Belief Networks-Classification by Backpropagation-Classification by Support Vector Machines-Lazy Learners.

UNIT – IV **[CO5, CO6]** **(12 Periods)**

Cluster Analysis: Introduction to cluster analysis- partitioning methods- Hierarchical methods- Density-Based Methods: DBSCAN, Outliers and Outlier Analysis- Outlier Detection Methods.

Data Mining Trends: Mining Sequence Data- Mining Graphs and Networks- Mining Other Kinds of Data- Data Mining Applications.

Learning Resources:

Text Books:

1. Data Mining Concepts & Techniques, Jiawei Han, Micheline Kamber, and Jian Pei, 3/e, Morgan Kaufmann Publishers.

Reference Books:

1. Introduction to Data Mining, Pang-Ning Tan, Michael Steinbach, and Vipin Kumar, Addison Wesley.
2. Data Warehouse Toolkit, Ralph Kimball, John Wiley Publishers.